

SQL Internship Project – Task 1

Basic Data Retrieval & Filtering

Final Project Title

Employee Data Retrieval and Filtering using SQL

Project Objective

This project demonstrates foundational SQL querying skills using SELECT statements, filtering, sorting, pattern matching, and logical operators.

The project is designed according to internship Task 1 requirements and can be uploaded directly to GitHub as an internship submission.

Technologies Used

- SQL
 - MySQL
 - MySQL Workbench
 - GitHub
-

Project Folder Structure

```
SQL-Task1-Data-Retrieval/  
├── dataset/  
│   └── employee_data.csv  
├── sql_queries/  
│   ├── create_table.sql  
│   ├── insert_data.sql  
│   ├── select_queries.sql  
│   ├── filtering_queries.sql  
│   ├── sorting_queries.sql  
│   └── pattern_matching.sql  
└── screenshots/
```

```
|   |─ table_output.png
|   |─ filtering_output.png
|   |─ sorting_output.png
|   |─ pattern_output.png
|
|─ README.md
|─ requirements.txt
```

Step 1 – Create Database

File: `create_table.sql`

```
CREATE DATABASE company_db;

USE company_db;

CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100),
    department VARCHAR(50),
    salary INT,
    city VARCHAR(50),
    joining_date DATE
);
```

Step 2 – Insert Sample Data

File: `insert_data.sql`

```
INSERT INTO employees VALUES
(1, 'Rahul Sharma', 'Sales', 50000, 'Delhi', '2023-01-10'),
(2, 'Priya Singh', 'HR', 45000, 'Mumbai', '2022-05-15'),
(3, 'Amit Verma', 'IT', 70000, 'Bangalore', '2021-08-20'),
(4, 'Sneha Kapoor', 'Sales', 55000, 'Delhi', '2023-03-12'),
(5, 'Rohan Gupta', 'Finance', 65000, 'Pune', '2020-09-05');
```

Step 3 – Column Selection Queries

File: `select_queries.sql`

```
-- Select all columns
SELECT * FROM employees;

-- Select specific columns
SELECT emp_name, department FROM employees;

-- Calculated column
SELECT emp_name, salary, salary * 1.1 AS new_salary
FROM employees;
```

Step 4 – Precision Filtering Queries

File: `filtering_queries.sql`

```
-- Equality operator
SELECT * FROM employees
WHERE department = 'Sales';

-- Salary greater than
SELECT * FROM employees
WHERE salary > 50000;

-- BETWEEN operator
SELECT * FROM employees
WHERE salary BETWEEN 45000 AND 65000;

-- IN operator
SELECT * FROM employees
WHERE city IN ('Delhi', 'Mumbai');

-- AND operator
SELECT * FROM employees
WHERE department = 'Sales' AND salary > 50000;

-- OR operator
SELECT * FROM employees
WHERE city = 'Delhi' OR city = 'Pune';

-- NOT operator
```

```
SELECT * FROM employees
WHERE NOT department = 'HR';
```

Step 5 – Sorting Queries

File: `sorting_queries.sql`

```
-- Sort by salary ascending
SELECT * FROM employees
ORDER BY salary ASC;

-- Sort by salary descending
SELECT * FROM employees
ORDER BY salary DESC;

-- Multiple column sorting
SELECT * FROM employees
ORDER BY department ASC, salary DESC;
```

Step 6 – Pattern Matching Queries

File: `pattern_matching.sql`

```
-- Names starting with R
SELECT * FROM employees
WHERE emp_name LIKE 'R%';

-- Names ending with a
SELECT * FROM employees
WHERE emp_name LIKE '%a';

-- Names containing 'it'
SELECT * FROM employees
WHERE department LIKE '%IT%';
```

README.md Content

```
# SQL Task 1 – Basic Data Retrieval & Filtering

## Project Overview
```

This project demonstrates basic SQL querying operations including:

- SELECT statements
- Filtering records
- Sorting data
- Pattern matching
- Logical operators

Tools Used

- MySQL
- SQL
- GitHub

Features

- Retrieve specific columns
- Apply WHERE conditions
- Use BETWEEN, IN, LIKE operators
- Sort data using ORDER BY
- Perform logical filtering with AND/OR/NOT

Author

Your Name

requirements.txt

MySQL Workbench
MySQL Server

Screenshots You Should Take

After running the SQL queries, take screenshots of:

1. Database creation
2. Table creation
3. Inserted records
4. Filtering query output
5. Sorting query output
6. LIKE operator output

Save them inside:

screenshots/

How to Run the Project

1. Open MySQL Workbench
 2. Create a new SQL file
 3. Run `create_table.sql`
 4. Run `insert_data.sql`
 5. Execute all query files one by one
 6. Save screenshots of outputs
-

GitHub Repository Setup

Repository Name

SQL-Task1-Data-Retrieval

Repository Description

SQL internship project demonstrating basic data retrieval and filtering operations.

Steps to Upload on GitHub

1. Open GitHub
2. Click "New Repository"
3. Repository Name:

SQL-Task1-Data-Retrieval

1. Select Public
 2. Add README
 3. Click Create Repository
-

Upload Project Files

Upload:

- dataset folder
- sql_queries folder
- screenshots folder

- README.md
- requirements.txt

Commit message:

Initial SQL internship project upload

Final GitHub Link Format

`https://github.com/yourusername/SQL-Task1-Data-Retrieval`

Example:

`https://github.com/aditya123/SQL-Task1-Data-Retrieval`

Internship Submission Checklist

Item	Status
SQL Files	✓
Dataset	✓
Screenshots	✓
README.md	✓
GitHub Repository	✓
Query Outputs	✓

Important Tips

- Use proper SQL formatting
 - Keep screenshots clean
 - Use comments in SQL queries
 - Upload all files properly
 - Keep repository public
 - Add meaningful commit messages
-

Recommended Commit Messages

```
Initial database setup
Added employee dataset
Implemented filtering queries
Added sorting operations
Updated README
Final internship submission
```